

Project LAB Data Modelling

Kelas: BL11

Kelompok: 3

Anggota:

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1. Data Cleaning and Preparation

a. Working With Numbers

- Pada sheet orders, ditemukan kolom days_since_prior_order yang memiliki *blank cells* (null).

order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order
2539329	1	prior	1	2	8	
2398795	1	prior	2	3	7	15
473747	1	prior	3	3	12	21
2254736	1	prior	4	4	7	29
431534	1	prior	5	4	15	28
3367565	1	prior	6	2	7	19
550135	1	prior	7	1	9	20
3108588	1	prior	8	1	14	14
2295261	1	prior	9	1	16	0
2550362	1	prior	10	4	8	30
1187899	1	train	11	4	8	14
2168274	2	prior	1	2	11	
1501582	2	prior	2	5	10	10
1901567	2	prior	3	1	10	3
738281	2	prior	4	2	10	8
1673511	2	prior	5	3	11	8
1199898	2	prior	6	2	9	13
3194192	2	prior	7	2	12	14
788338	2	prior	8	1	15	27
1718559	2	prior	9	2	9	8
1447487	2	prior	10	1	11	6
1402090	2	prior	11	1	10	30
3186735	2	prior	12	1	9	28

- Menerapkan Working with Numbers dengan mengisi nilai kosong (null) menjadi 0.

×

Replace Values

Replace one value with another in the selected columns.

Value To Find

null

Replace With

0

OK

Cancel

APPLIED STEPS

Source	⚙
Promoted Headers	⚙
Changed Type	
✕ Replaced Value	⚙

order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order
2539329	1	prior	1	2	8	0
2398795	1	prior	2	3	7	150
473747	1	prior	3	3	12	210
2254736	1	prior	4	4	7	290
431534	1	prior	5	4	15	280
3367565	1	prior	6	2	7	190
550135	1	prior	7	1	9	200
3108588	1	prior	8	1	14	140
2295261	1	prior	9	1	16	0
2550362	1	prior	10	4	8	300
1187899	1	train	11	4	8	140
2168274	2	prior	1	2	11	0
1501582	2	prior	2	5	10	100
1901567	2	prior	3	1	10	30
738281	2	prior	4	2	10	80
1673511	2	prior	5	3	11	80
1199898	2	prior	6	2	9	130
3194192	2	prior	7	2	12	140
788338	2	prior	8	1	15	270
1718559	2	prior	9	2	9	80
1447487	2	prior	10	1	11	60
1402090	2	prior	11	1	10	300
3186735	2	prior	12	1	9	280
3268552	2	prior	13	4	11	300
839880	2	prior	14	3	10	130
1492625	2	train	15	1	11	300
1374495	3	prior	1	1	14	0
444309	3	prior	2	3	19	90
3002854	3	prior	3	3	16	210

b. Manipulating Text

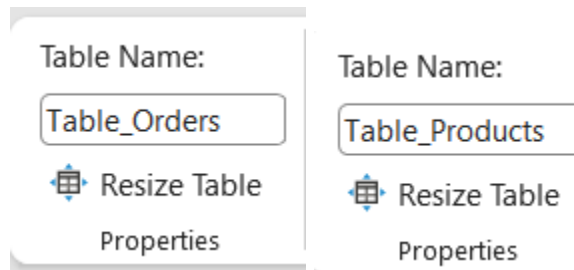
- Membuat kolom baru ID_Prefix untuk mengelompokkan data.
- Menggunakan fungsi LEFT untuk mengambil 3 angka pertama dari kolom order_id.
- Mempermudah pengelompokan data berdasarkan kategori wilayah atau jenis transaksi saat analisis nanti.

H2								
=LEFT(A2; 3)								
	A	B	C	D	E	F	G	H
1	order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order	ID_Prefix
2	2539329	1	prior	1	2	8	0	253
3	2398795	1	prior	2	3	7	150	239
4	473747	1	prior	3	3	12	210	473
5	2254736	1	prior	4	4	7	290	225
6	431534	1	prior	5	4	15	280	431
7	3367565	1	prior	6	2	7	190	336
8	550135	1	prior	7	1	9	200	550
9	3108588	1	prior	8	1	14	140	310
10	2295261	1	prior	9	1	16	0	229
11	2550362	1	prior	10	4	8	300	255
12	1187899	1	train	11	4	8	140	118
13	2168274	2	prior	1	2	11	0	216
14	1501582	2	prior	2	5	10	100	150
15	1901567	2	prior	3	1	10	30	190
16	738281	2	prior	4	2	10	80	738
17	1673511	2	prior	5	3	11	80	167
18	1199898	2	prior	6	2	9	130	119
19	3194192	2	prior	7	2	12	140	319
20	788338	2	prior	8	1	15	270	788
21	1718559	2	prior	9	2	9	80	171
22	1447487	2	prior	10	1	11	60	144
23	1402090	2	prior	11	1	10	300	140
24	3186735	2	prior	12	1	9	280	318
25	3268552	2	prior	13	4	11	300	326
26	839880	2	prior	14	3	10	130	839
27	1492625	2	train	15	1	11	300	149
28	1374495	3	prior	1	1	14	0	137
29	444309	3	prior	2	3	19	90	444
30	3002854	3	prior	3	3	16	210	300

2. Named Ranges and Tables

a. Excel Table Creation

- Mengonversi dataset orders dan products menjadi *Excel Table* agar memiliki struktur yang dinamis.
- Table_Orders dan Table_Products.
- Mempermudah referensi data dalam rumus serta memastikan sinkronisasi otomatis saat ada penambahan data baru.



b. Named Ranges

Pembuatan *Named Ranges* dilakukan untuk menyederhanakan proses analisis dan penggabungan data antar tabel:

- List_ProductID: Referensi untuk kolom product_id pada Table_Products.
- List_ProductName: Referensi untuk kolom product_name pada Table_Products.

productName	:	fx	product_name
List_ProductID			B
List_ProductN..			product_name
aisles	1		Chocolate Sandwich Cookies
departments	2		All-Seasons Salt
order_product..	3		Robust Golden Unsweetened Oolong Tea
order_product..	4		Smart Ones Classic Favorites Mini Rigatoni With Vodka Cream Sauce
orders	5		Green Chile Anytime Sauce
products	6		Dry Nose Oil
Table_Orders	7		Pure Coconut Water With Orange
Table_Products	8		Cut Russet Potatoes Steam N' Mash
10	9		Light Strawberry Blueberry Yogurt
11	10		Sparkling Orange Juice & Prickly Pear Beverage
12	11		Peach Mango Juice
13	12		Chocolate Fudge Layer Cake
14	13		Saline Nasal Mist
15	14		Fresh Scent Dishwasher Cleaner
16	15		Overnight Diapers Size 6
17	16		Mint Chocolate Flavored Syrup
18	17		Rendered Duck Fat
19	18		Pizza for One Suprema Frozen Pizza
20	19		Gluten Free Quinoa Three Cheese & Mushroom Blend
21	20		Pomegranate Cranberry & Aloe Vera Enrich Drink
22	21		Small & Medium Dental Dog Treats
23	22		Fresh Breath Oral Rinse Mild Mint
24	23		Organic Turkey Burgers
25	24		Tri-Vi-Sol® Vitamins A-C-and D Supplement Drops for Infants

3. Logical, Lookup, and Conditional Functions

a. Logical Functions

- Melakukan transformasi data numerik pada kolom order_dow menjadi nama hari yang informatif.
- Menggunakan fungsi *IF* untuk memetakan nilai numerik (1-7) menjadi nama hari (Sunday-Saturday).
- Mengonversi data mentah menjadi informasi yang lebih mudah diinterpretasikan untuk analisis pola transaksi pelanggan berdasarkan hari.

=IF(E2=1; "Sunday"; IF(E2=2; "Monday"; IF(E2=3; "Tuesday"; IF(E2=4; "Wednesday"; IF(E2=5; "Thursday"; IF(E2=6; "Friday"; "Saturday"))))))																	
1	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q
order_id	user_id	eval_set	order_number	order_dow	order_hour_of_day	days_since_prior_order	ID Prefix	Day Name									
2	2539329	1 prior	1	2	8	0	253	Monday									
3	2398795	1 prior	2	3	7	150	239	Tuesday									
4	473747	1 prior	3	3	12	210	473	Tuesday									
5	2254736	1 prior	4	4	7	290	225	Wednesday									
6	431534	1 prior	5	4	15	280	431	Wednesday									
7	3367565	1 prior	6	2	7	190	336	Monday									
8	550135	1 prior	7	1	9	200	550	Sunday									
9	3108588	1 prior	8	1	14	140	310	Sunday									
10	2295261	1 prior	9	1	16	0	229	Sunday									
11	2550362	1 prior	10	4	8	300	255	Wednesday									
12	1187899	1 train	11	4	8	140	118	Wednesday									
13	2168274	2 prior	1	2	11	0	216	Monday									
14	1501582	2 prior	2	5	10	100	150	Thursday									
15	1901567	2 prior	3	1	10	30	190	Sunday									
16	738281	2 prior	4	2	10	80	738	Monday									
17	1673511	2 prior	5	3	11	80	167	Tuesday									
18	1198988	2 prior	6	2	9	130	119	Monday									
19	3194192	2 prior	7	2	12	140	319	Monday									
20	788338	2 prior	8	1	15	270	788	Sunday									
21	1718559	2 prior	9	2	9	80	171	Monday									
22	1447487	2 prior	10	1	11	60	144	Sunday									
23	1402090	2 prior	11	1	10	300	140	Sunday									
24	3186735	2 prior	12	1	9	280	318	Sunday									
25	3268552	2 prior	13	4	11	300	326	Wednesday									
26	R39880	2 prior	14	3	10	130	R39	Tuesday									

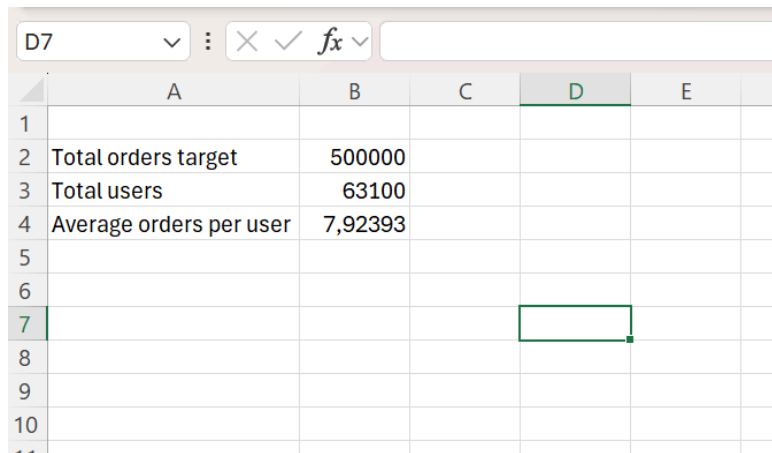
b. Conditional Functions

- Melakukan ringkasan (Data Summarization) untuk distribusi produk per departemen.
- Menggunakan fungsi COUNTIF pada tabel departments untuk menghitung frekuensi kemunculan setiap department_id di dalam tabel products.
- Mendapatkan ringkasan yang akurat mengenai jumlah produk yang tersedia dalam setiap departemen untuk mendukung pengambilan keputusan.

=COUNTIF(Table_Deptamen; [@[department_id]])									
1	department_id	department	Total_Products						
2	1	frozen	4007						
3	2	other	548						
4	3	bakery	1516						
5	4	produce	1684						
6	5	alcohol	1054						
7	6	international	1139						
8	7	beverages	4365						
9	8	pets	972						
10	9	drygoods pasta	1858						
11	10	bulk	38						
12	11	personal care	6563						
13	12	meat seafood	907						
14	13	pantry	5371						
15	14	breakfast	1115						
16	15	canned goods	2092						
17	16	dairy eggs	3449						
18	17	household	3085						
19	18	babies	1081						
20	19	snacks	6264						
21	20	deli	1322						
22	21	missing	1258						

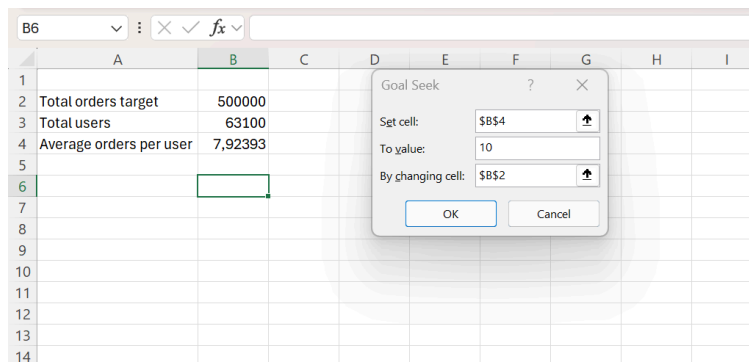
4. What If Analysis (Goal Seek)

- Mencari rata-rata order per user jika total order mencapai target tertentu
- Mengawali dengan membuat tabel seperti pada gambar dibawah ini
- Menggunakan rumus pada Total orders target : diisi manual dengan angka target order yaitu 500000, kemudian **rumus total users**
:=SUMPRODUCT(1/COUNTIF(orders[user_id],orders[user_id])) dan **Average orders per user** yaitu =B2/B3



	A	B	C	D	E
1					
2	Total orders target	500000			
3	Total users	63100			
4	Average orders per user	7,92393			
5					
6					
7					
8					
9					
10					

- Setelah itu masuk ke Data - What if analysis - Goal seek
- Kemudian isi pada bagian set cell, to value dan by changing cell
- Kemudian pencet OK

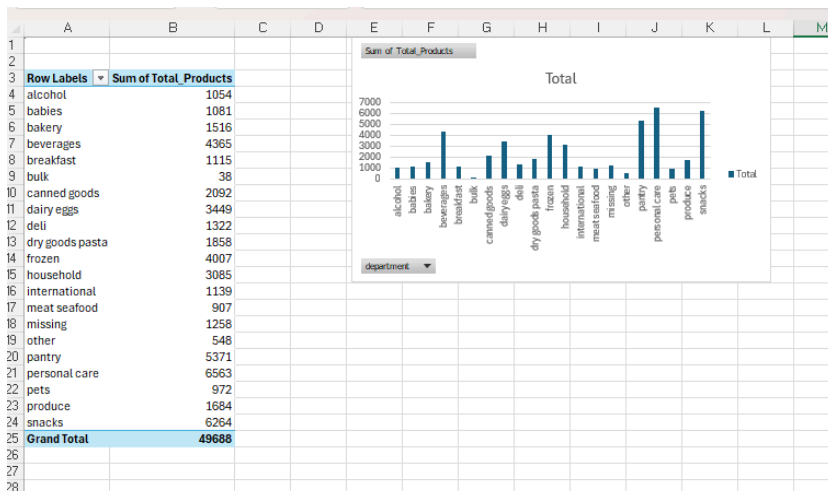


Calculated Fields

- Kemudian, membuat calculated measure menggunakan DAX. Measure yang digunakan yaitu **Total_Products : =COUNTA(order_products__train[product_id])** dan **Total Orders:=COUNT(orders[order_id])**

[order_... - fx]				
	ord...	user_id	eval_set	order_r
1	2539329		1 prior	
2	2398795		1 prior	
3	473747		1 prior	
4	2254736		1 prior	
5	431534		1 prior	
6	3367565		1 prior	
7	550135		1 prior	
8	3108588		1 prior	
9	2295261		1 prior	
10	2550362		1 prior	
11	1187899		1 train	
12	2168274		2 prior	
13	1501582		2 prior	
14	1901567		2 prior	
15	738281		2 prior	

		Total Orders	Measure 1: 1048575
		Total Products So...	Measure 2: 1048575



- Cube Value

`=CUBEVALUE("ThisWorkbookDataModel", "[Measures].[Sum of Total_Products]")`

The screenshot shows an Excel spreadsheet with the formula `=CUBEVALUE("ThisWorkbookDataModel", "[Measures].[Sum of Total_Products]")` entered in cell E2. The result of the formula, 49688, is displayed in the cell. The spreadsheet has columns labeled D through O and rows 1 through 10. A 'Help' button is visible on the right side of the interface.

6. Conditional and Number Formatting

- Conditional Formatting

Highlights bagian reordered dengan no 1 yaitu green fill dan 0 red fill

P12

	A	B	C	D	E	F	G
1	order_id	product_id	add_to_cart_order	reordered			
2	1	49302	1	1			
3	1	11109	2	1			
4	1	10246	3	0			
5	1	49683	4	0			
6	1	43633	5	1			
7	1	13176	6	0			
8	1	47209	7	0			
9	1	22035	8	1			
10	36	39612	1	0			
11	36	19660	2	1			
12	36	49235	3	0			
13	36	43086	4	1			
14	36	46620	5	1			
15	36	34497	6	1			
16	36	48679	7	1			
17	36	46979	8	1			
18	38	11913	1	0			
19	38	18159	2	0			
20	38	4461	3	0			
21	38	21616	4	1			
22	38	23622	5	0			
23	38	32433	6	0			
24	38	28842	7	0			
25	38	42625	8	0			

order_products_train order_products_prior Sheet4

Ready Accessibility: Investigate

-Number formatting

Mengubah tampilan angka supaya lebih rapih dan mudah di baca.

Cara nya yaitu : pilih salah satu angka di bawah -> klik kanan -> value field settings -> number format -> number -> checklist bagian use 1000 separator -> decimal placesnya 2 -> oke



	A	B	C
1			
2			
3	Row Labels	Sum of Total Products	
4	alcohol	1.054,00	
5	babies	1.081,00	
6	bakery	1.516,00	
7	beverages	4.365,00	
8	breakfast	1.115,00	
9	bulk	38,00	
0	canned goods	2.092,00	
1	dairy eggs	3.449,00	
2	deli	1.322,00	
3	dry goods pasta	1.858,00	
4	frozen	4.007,00	
5	household	3.085,00	
6	international	1.139,00	
7	meat seafood	907,00	
8	missing	1.258,00	
9	other	548,00	
0	pantry	5.371,00	
1	personal care	6.563,00	
2	pets	972,00	
3	produce	1.684,00	
4	snacks	6.264,00	
5	Grand Total	49.688,00	
6			
7			

- Calculated Fields and Measures

Dibuat menggunakan DAX pada Power Pivot dengan rumus :

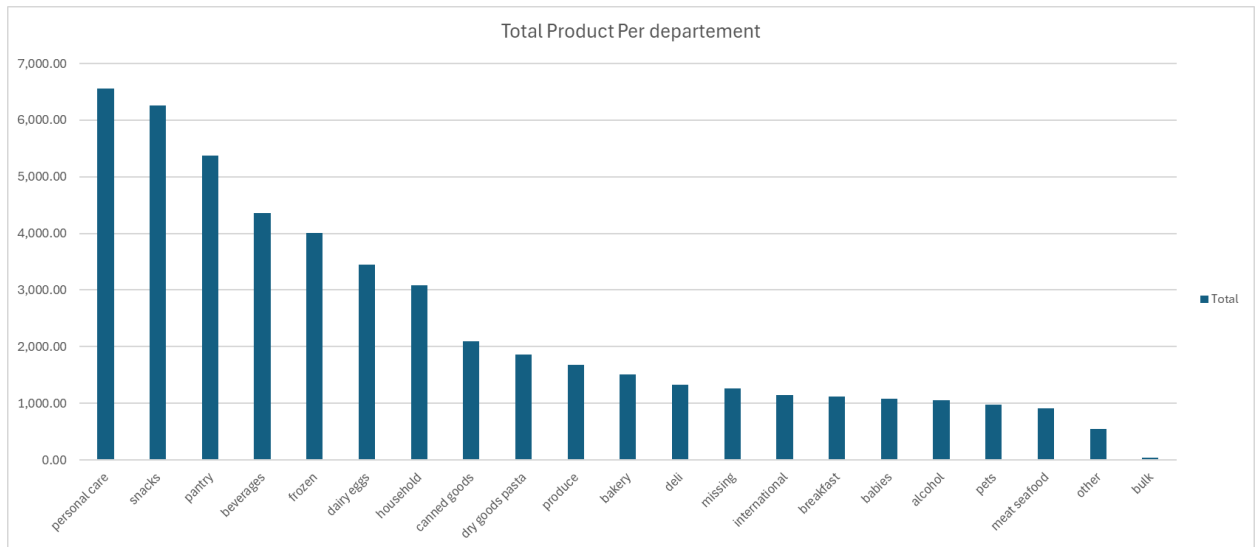
```
Average_Reorder:=AVERAGE(order_products__train[reordered])
```

(measure tersebut digunakan untuk menghitung total product berdasarkan kolom reordered

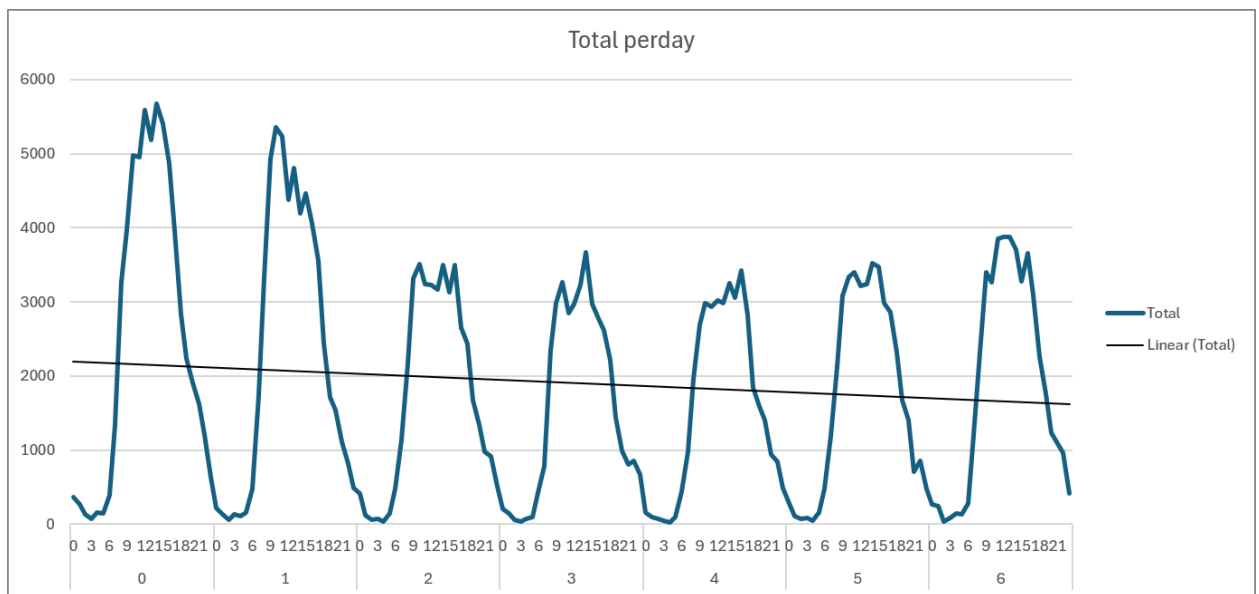
[eval_s...		fx Measure 2:=COUNT(order_products__train[product_id])	
ord...	user_id	eval_set	order_nu...
1	2539329	1 prior	
2	2398795	1 prior	
3	473747	1 prior	
4	2254736	1 prior	
5	431534	1 prior	
6	3367565	1 prior	
7	550135	1 prior	
8	3108588	1 prior	
9	2295261	1 prior	
10	2550362	1 prior	
11	1187899	1 train	
12	2168274	2 prior	
13	1501582	2 prior	
14	1901567	2 prior	
15	738281	2 prior	
16	1673511	2 prior	
17	1199898	2 prior	
18	3194192	2 prior	
	Total Orders	Measure 1: 1048575	
	Total Products So...	Measure 2: 1048575	

7. Mastering Chart

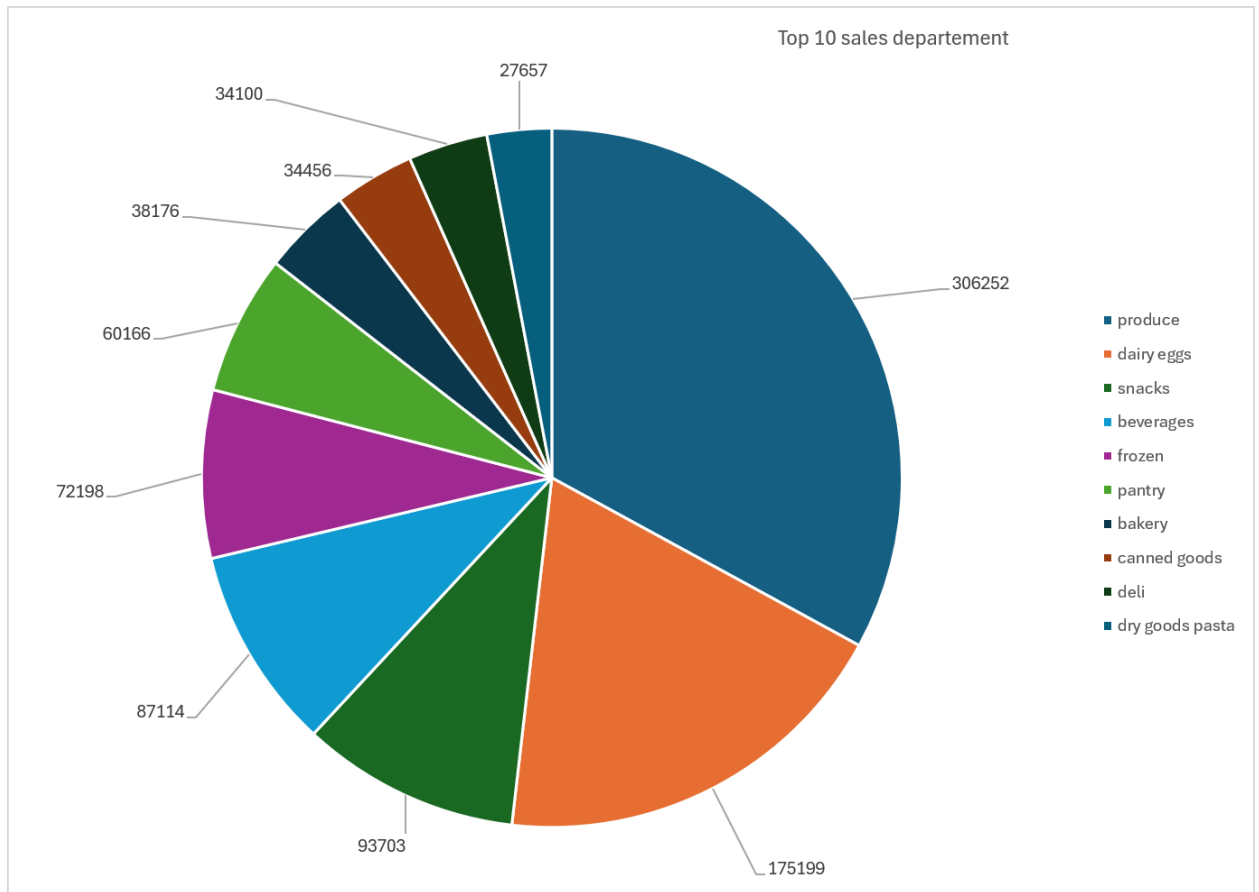
- a. Bar chart : menampilkan total Product di masing masing departemen untuk melihat distribusi product di masing masing departemen



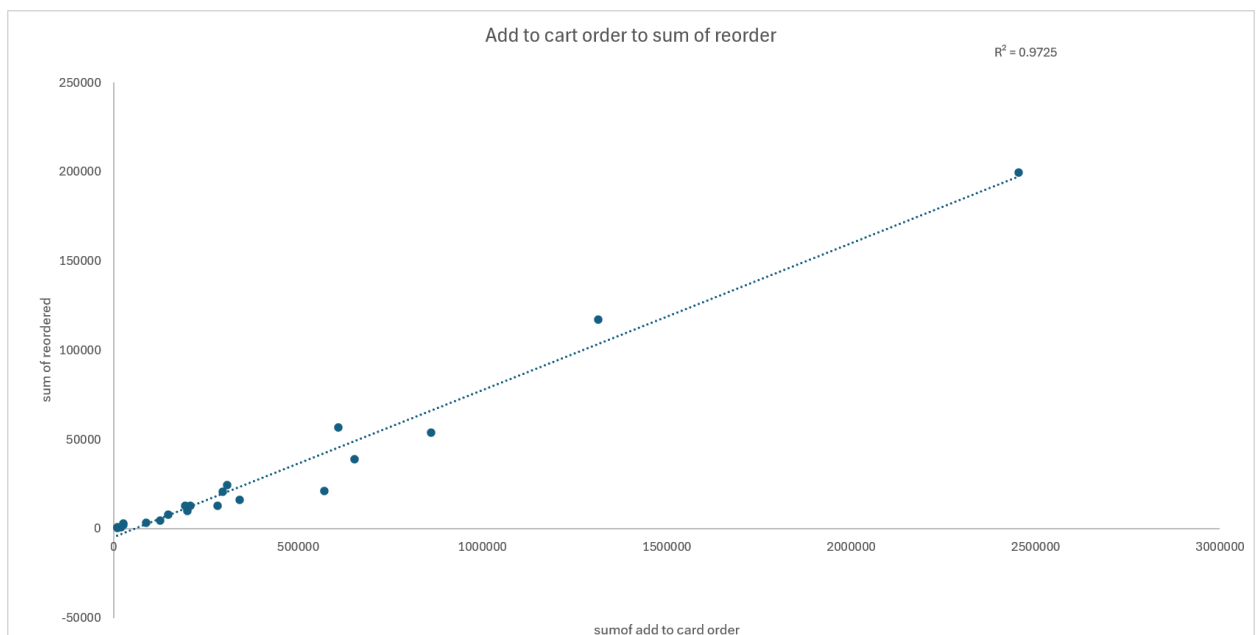
- b. Line chart : menampilkan timeline total product dalam hari yang dibagi dalam jam perhari dalam satu minggu. Chart ini untuk mengekstrak informasi mengenai kapan penjualan menaik di jam / hari tertentu



- c. Pie Chart : Menampilkan total sales per departement. Chart ini dapat memberikan informasi departement mana yang paling banyak terjual

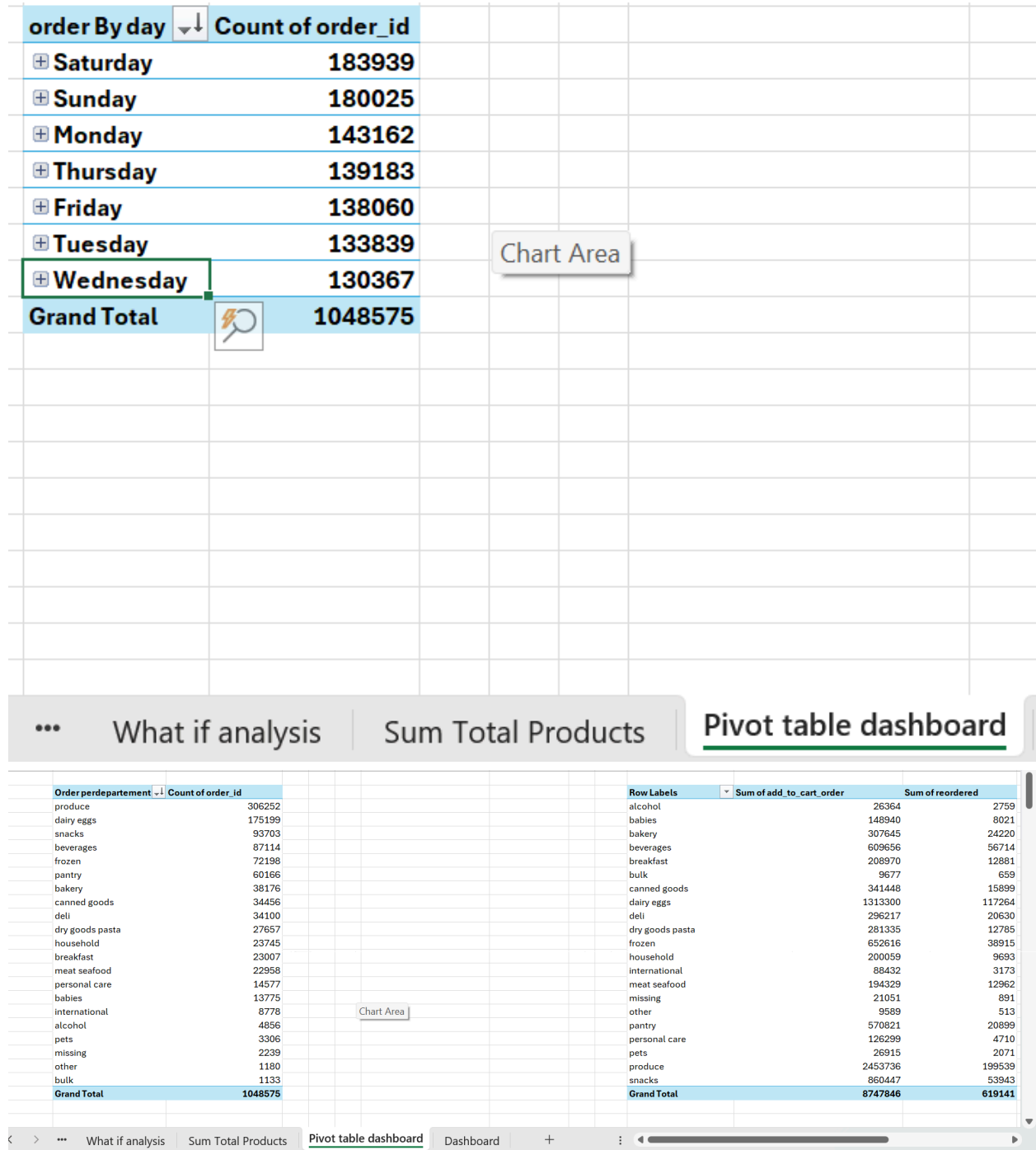


- d. Scatter plot : membandingkan antara add_to_cart_order dengan Reordered product.
Semakin banyak add_to_cart_order maka semakin banyak Product yang Reordered



8. Interactive Dashboard

Sebelumnya dibuat sheet yang mengumpulkan beberapa pivot table untuk dibuat menjadi interaktif chart agar mempermudah dalam pembuatannya



Dalam dashboard ada 2 chart yang interaktif yaitu Pie chart dan Line chart serta 2 slicer yang digunakan.

1. Slicer days: dalam slicer ini menampilkan data total order product by departemen per harinya
2. Slicer Reorder vs order : menampilkan jumlah product apa saja yang di order ataupun reorder

Link Excel:

https://binusianorg-my.sharepoint.com/personal/iman_damsyai_binus_ac_id/_layouts/15/guestaccess.aspx?share=IQBAAt0fojGHiTLA_75S-VHaqAU3AcDxNQlhw0IJJa6sprbGw&e=UitDPz